# Growth applications

**1**

**a**

**i** Exponential growth **ii** $60\%$

**b**

**i** Exponential decay **ii** $80\%$

**c**

**i** Exponential growth **ii** $105\%$

**d**

**i** Exponential decay **ii** $22\%$

**2** $155\,432$

**3** $0.70$ AU

**4**

**a** $1000$ aphids **b** $32\,000$ aphids **c** $250$ aphids

**5**

**a** $8$ million

**b**

**i** $7130$ million **ii** $9286$ million

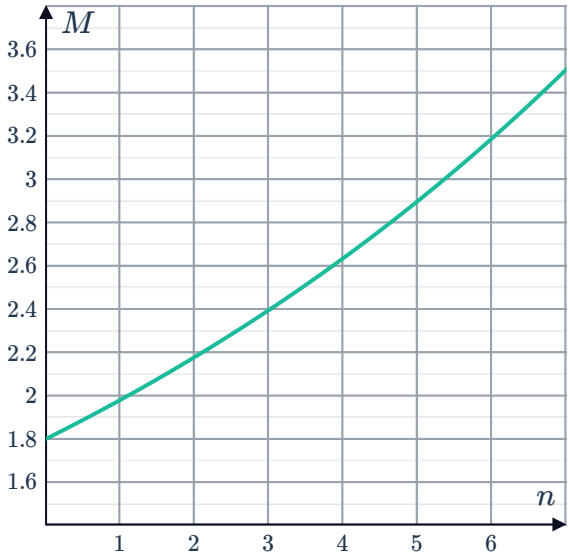
**6**

**a** $M = 2.4$

**b**

| Months ($n$) | 0 | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|---|
| Mass ($M$) | 1.8 | 2.0 | 2.2 | 2.4 | 2.6 | 2.9 | 3.2 |

**c**




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**7**

**a**

| $x$ | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| $y$ | 2 | 4 | 8 | 16 |

**b** 2

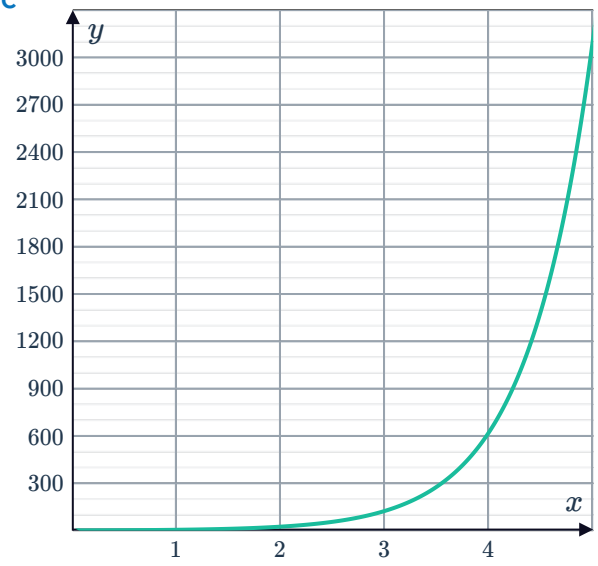
**c** $y = 2^x$

**8**

**a** No

**b** $y = (5)^x$

**c**




**9**

**a** 5

**b** $y = -6 \times 5^x$

**10**

**a**

| No. of folds $(f)$ | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| No. of rectangles $(r)$ | 1 | 2 | 4 | 8 |

**b** 2

**c** $r = 2^f$

**d** 2048

**11**

**a** $y = 10 \times 3^x$

**b** $y = 200 \times 1.25^x$

**12**

**a** $y = 750\,(1.03)^t$

**b** 869 rats

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**13**

**a** $\frac{5}{6}$

**b** $f(t) = \frac{5}{6}(1+1)^t$

**c** Growth

**d** $100\%$

**14**

**a** $A = 5100(1.112)^y$

**b** $11.2\%$

**15**

**a** $r = e^{0.09} - 1$

**b** $A = 2200(1 + 0.0942)^x$

**c** $9.42\%$

**16**

**a**

| Seconds passed $(t)$ | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| Number of bacteria $(y)$ | 1 | 2 | 4 | 8 | 16 | 32 |

**b** $y = 2^t$

**c** $2^{60}$

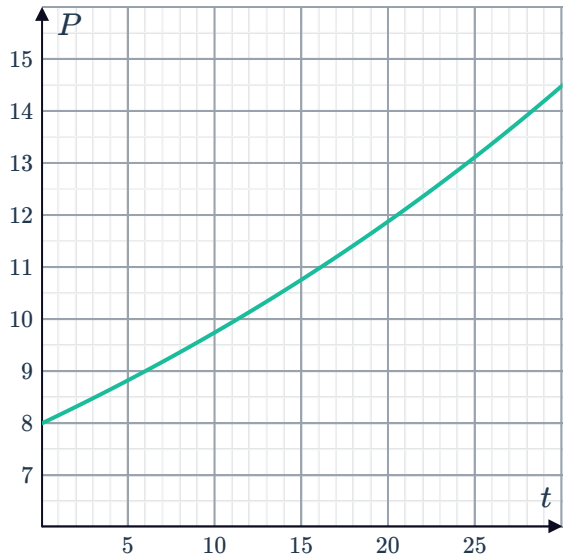
**d** $y = 2^{3600h}$

**17**

**a**

| $t$ | 5 | 10 | 15 | 20 | 25 | 30 |
|---|---|---|---|---|---|---|
| $P$ | 8.8 | 9.8 | 10.8 | 11.9 | 13.1 | 14.5 |

**b**




**c** 23 years

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**18**

**a** $y = 160\,(1.19)^t$

**b** $n = 9$

**19**

**a** $P = 90 + 5t$

**b** $Q = 90 \times (1.04)^t$

**c** The current company

**d** Opening his own business

**20**

**a** Treatment A

**b** $5$

**c** Treatment A

**d** $72\,550$

**21**

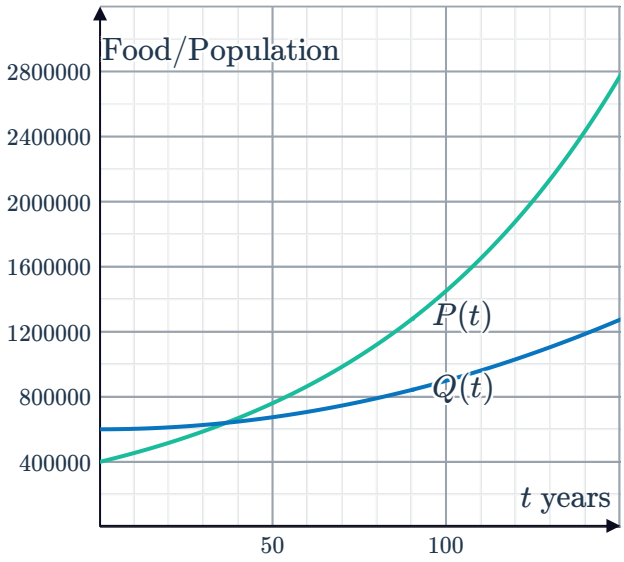
**a** $147\,850$

**b** There will be a surplus of food in $10$ years time.

**c** $-555\,474$

**d** There will be a food shortage in $100$ years time.

**e**




**f** $36.34$ years

**g** No, the population increases at a faster rate than the food supply. Once the population has overtaken it will stay above and there will be an increasing gap.

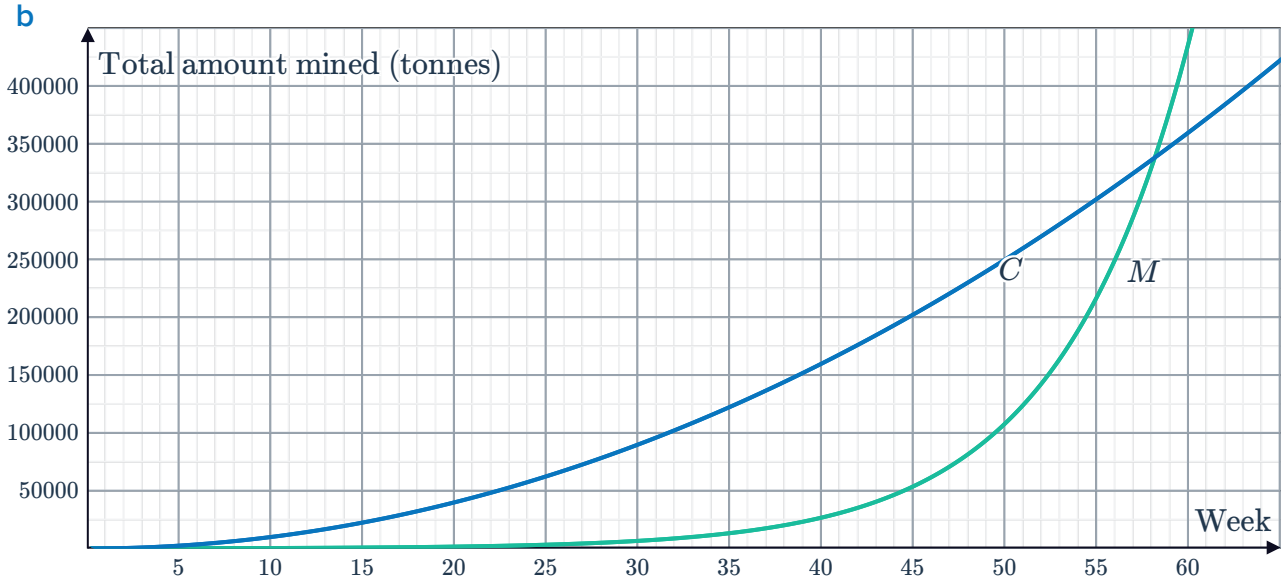
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**22**

**a**

| $n$ | $M$ | $C$ |
|---|---|---|
| 1 | 100 | 100 |
| 5 | 175 | 352 |
| 10 | 708 | 2500 |
| 15 | 10 000 | 22 500 |

**b**




**c** Crest Corporation

**d** $(1, 100)$ and $(59.5, 353563.1)$, at the end of the first week of operations the two companies have mined the same total of 100 tonnes. The Crest Corporation then has a higher total amount mined until the approximately 59.5 weeks later when they are equal again at $353563.1$ tonnes mined in total.

**e** Mint Corporation

## Decay applications

**23**

**a** $y = 840\,(0.96)^m$

**b** 606 rodents

**24**

**a** $y = 2310\,(0.974)^n$

**b** 1328

**25**

**a** $y = 21\,(0.5)^n$

**b** 0.66 m

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**26**

**a**

i $179$ ii $132$

**b**

i $70\%$ ii $40\%$

**c** If you have a high sugar diet, your blood sugar level peaks at higher amounts and decreases more rapidly, than for a person with a low sugar diet.

**27**

**a** Decreasing

**b** i No ii Yes iii Yes iv No

**c** The initial temperature of the substance

**d** $T = 145 \times 0.6^t$

**e** $0.88\,^\circ\text{C}$

**f** No

**g** Substance $P$

**28**

**a** $a = 5$ **b** $b = 4$ **c** $y = \dfrac{5}{16}$

**29**

**a** $y = 10 \times 3^x$ **b** $y = 200 \times 1.25^x$

**30**

**a** $2$

**b** $f(t) = 2(1 - 0.625)^t$

**c** Decay

**d** $62.5\%$

**31**

**a** $A = 60(0.5)^{\frac{n}{8}}$ **b** $0.12$ g

**32**

**a** $A = 300(0.5)^{\frac{n}{20}}$ **b** $0.59$ g

**33**

**a** $A = 385(0.5)^{\frac{n}{5730}}$ **b** $321.11$ g **c** $19\,823$ years

**34**

**a** $A = k \times (1 - 0.0245)^t$ **b** $2.45\%$

**35**

**a** $A = 8900(1 - 0.0769)^t$ **b** $7.69\%$

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**36**

a $A = 3400\,(1 - 0.2054)^t$

b $20.54\%$

**37**

a $P = (0.85)^t$

b Maximilian

c C

d $13\%$

e No

**38**

a $192.372$ acres

b $185.03$ acres

c $f\,(n) = 200\,(0.96186)^{n-1}$

d $3.814\%$

## Financial applications

**39**

a $y = 2900\,(1.09)^t$

b $\$5778.43$

**40** $3313.46$

**41** $241.70$

**42**

a $\$\,(1.1)^t$

b $\$\left(1 + \dfrac{0.1}{12}\right)^{12t}$

**43**

a

i $A = 6100\,(1.083)^t$

ii $\$9088.08$

b

i $A = 6800\,(1.0025)^{12t}$

ii $\$7898.99$

c

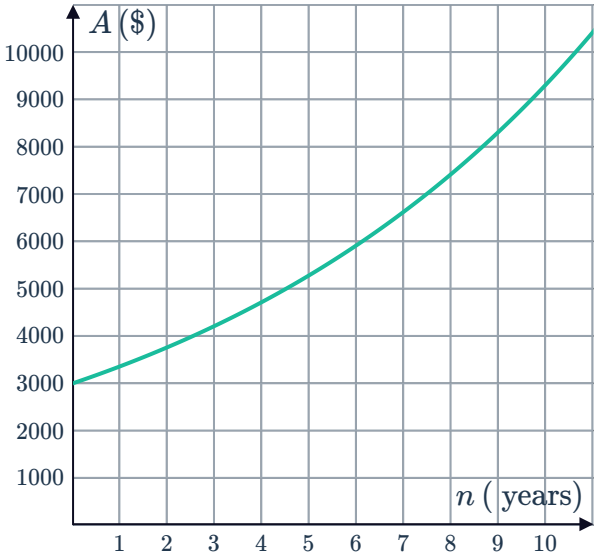
i $A = 3200\,(1.0013)^{4t}$

ii $\$4143.23$

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**44**

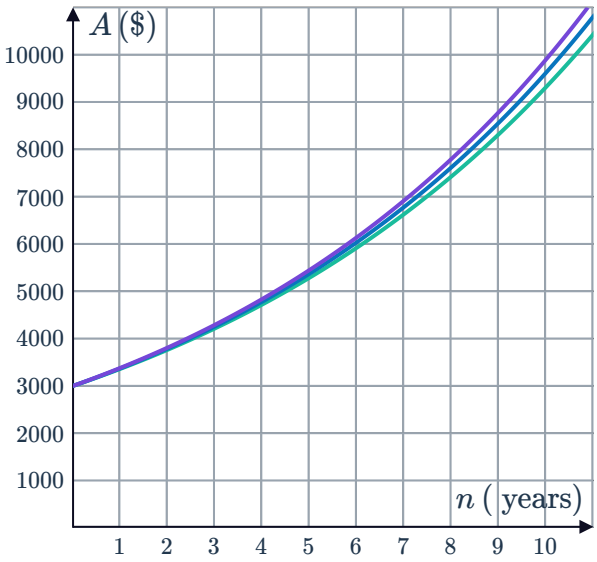
**a**




**b** $A = 3000 \times 1.06^{2n}$

**c** $A = 3000 \times 1.01^{12n}$

**d**




**e** $\$583.62$

**45**

**a** $0.02$

**b** $A = 1000 \times 1.02^{n}$

**c** $\$1126.16$

**46**

**a** $M = 50\,000\,(1.032)^{t}$

**b** $P = 49\,000\,(1.061)^{t}$

**c** $\$9500.24$

**d** Elizabeth's

**47**

**a** Increasing by a factor of 2 each day.

**b** $\$64$

**c** $y = 4\left(2^{x-1}\right)$

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**48**

**a**

| $x$ | 0 | 3 | 6 | 9 |
|---|---|---|---|---|
| Company A's bill | 200 | 218 | 236 | 254 |
| Company B's bill | 200 | 224.97 | 253.06 | 284.66 |

**b** Company B

**49**

**a** $f(n) = 125\,000 + 3000n$

**b** $g(n) = 20\,000\,(1.4)^n$

**c**

| $n$ | 2 | 3 | 4 |
|---|---|---|---|
| Increase in salary compared to previous year when working for a company | 3000 | 3000 | 3000 |
| Increase in salary compared to previous year when working for her own business | 11 200 | 15 680 | 21 952 |

**d** Working for a company

**e** Starting her own business

**50**

**a**

**i** $A = P\,(1.053)^t$

**ii** $A = P\,(1.0125)^{4t}$

**b** $5.09\%$

**c** Bank A

**51**

**a** $A = 17\,000\,(1 + 0.0444)^{4n}$

**b** $4.44\%$

**52**

**a**

**i** $A = 600\,(1.0046)^{12t}$

**ii** $A = 600\,(1.048)^t$

**b** $600\,(1 + 0.0039)^{12t}$

**c** $0.39\%$

**d** Savings account

**53**

**a** $y = 322\,000\,(0.95)^m$

**b** $\$249\,157.46$

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**54** **a** Sale price of property **b** $5.3\%$

**c** $-5.3\%$

**55** $5.08\%$

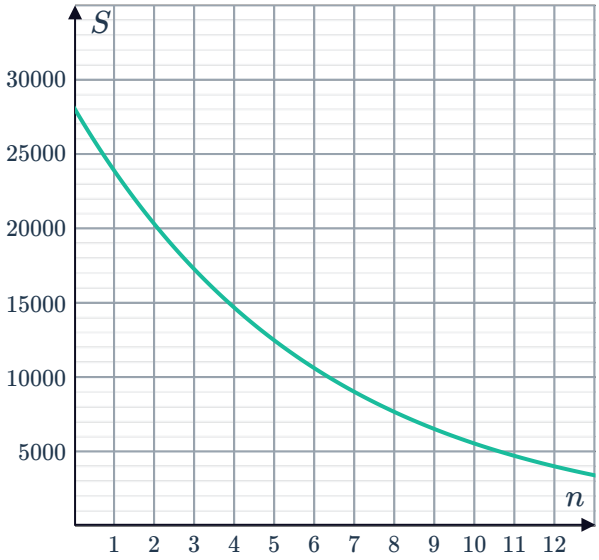
**56** **a** $f(t) = 1.27\,(0.92)^t$ **b** $8\%$

**c** $\$0.47$

**57** **a**

| $n$ | 2 | 4 | 6 | 8 | 10 | 12 |
|---|---|---|---|---|---|---|
| $S$ | 20 230 | 14 616.18 | 10 560.19 | 7629.73 | 5512.48 | 3982.77 |

**b**




**c** $\$17\,195.5$

**d** 9 years

**58** **a** $A = 45\,000\,(0.78)^t$ **b** $\$16\,656.78$

**59** **a** $0.01$ **b** $A = 38\,200 \times 0.99^n$

**c** $\$35\,964.54$ **d** Motorbike

**60** **a** $A = 25\,000\,(1 - 0.0031)^{52t}$ **b** $0.31\%$

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